Lecture 1: Foundations of Quantum Physics

Quantum mechanics arose in the early twentieth century to describe phenomena at microscopic scales where classical Newtonian mechanics fails.

Wave-particle duality states that matter and light exhibit behaviors characteristic of both waves and discrete particles.

Max Planck solved the ultraviolet catastrophe in black-body radiation by postulating that electromagnetic energy is quantized in discrete packets called quanta: E = h * nu.

Albert Einstein explained the photoelectric effect in 1905 by demonstrating that light consists of localized photons carrying energy proportional to frequency.

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Lecture 2: The Schrödinger Wave Equation

The state of a quantum physical system is completely described by its complex wave function Psi(r, t).

The time-dependent Schrödinger equation describes how the quantum state of a non-relativistic physical system changes over time: i * h_bar * d(Psi)/dt = H * Psi, where H is the Hamiltonian operator.

Born's statistical interpretation states that the square of the absolute value of the wave function, |Psi(r, t)|^2, gives the probability density of finding the particle at position r at time t.

Wave functions must satisfy mathematical boundary conditions including square-integrability, continuity, and single-valuedness across physical space.

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Lecture 3: Heisenberg Uncertainty Principle

Formulated by Werner Heisenberg in 1927, the uncertainty principle establishes a fundamental limit to the precision with which certain pairs of physical properties can be known simultaneously.

The position-momentum uncertainty relation is Delta(x) * Delta(p) >= h_bar / 2, indicating that position and linear momentum are canonically conjugate observables.

This principle is an intrinsic wave property of quantum systems, not merely a limitation of experimental measuring instruments.

Similarly, the energy-time uncertainty relation Delta(E) * Delta(t) >= h_bar / 2 governs the lifetime of excited quantum states and particle resonance widths.

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Lecture 4: Quantum Entanglement and Non-Locality

Quantum entanglement occurs when two or more particles become interconnected such that the quantum state of each particle cannot be described independently of the state of the others.

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen proposed the EPR paradox, questioning whether quantum mechanics is a complete theory or whether hidden variables exist.

In 1964, John Stewart Bell formulated Bell's theorem and Bell inequalities, proving that no local hidden-variable theory can reproduce all predictions of quantum mechanics.

Experimental verifications of Bell inequality violations have laid the foundation for modern quantum computing, quantum key distribution (QKD), and quantum teleportation protocols.

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